SUNWAY UNIVERSITY · SCHOOL OF MATHEMATICAL SCIENCES & COMPUTING

Autonomous Wildfire Risk Assessment & UAV Surveillance

Mamdani fuzzy inference driving a four-probe UAV inside an NVIDIA Isaac Sim digital twin

CSC3034 Computational Intelligence · Supervisor: Dr. Tang Tiong Yew · SDG 13 Climate Action · SDG 15 Life on Land

Yeoh Choon Hern
23100795
Project lead — coordination and integration

Pang Wei Sheng
24035115
Fuzzy inference engine & simulation evaluation

Roshan A/L Sivakumar
23028285
Technical report — defuzzification analysis

Dinesswar A/L Sathasivam
23096910
Isaac Sim OpenUSD stage — sensor bridge

NVIDIA Isaac Sim 6.2.0 · OpenUSD procedural stage · 4 inputs · 16 sets · 10 rules · 0.078 ms median inference · p99 0.36 ms

01 Why a threshold fails

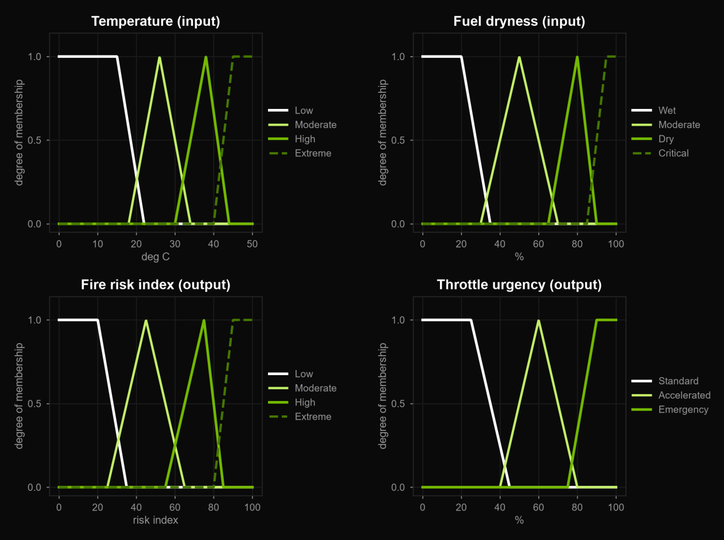
A single cut-off cannot separate **38 °C at 15 % humidity with 70 km/h wind** from **42 °C in saturated still air**. The first case is the dangerous one, and a temperature-only rule ranks it safe. Spread is non-linear in the same way: dry wind at moderate temperature ignites faster than a hot, humid afternoon.

BEHAVIOUR	THRESHOLD	MAMDANI
Decision surface	step at 40 °C	**continuous**
Humidity / wind	ignored	**fused**
Sensor noise	chatters at the edge	**averaged**
Output	alarm on / off	**throttle 20–90 %**

02 Linguistic variables

Four inputs, four sets each — 16 in total. Trapezoids close the outer sets, so a reading past the sensor range saturates instead of running off the universe.

VAR	UNIVERSE	SETS
T	0–50 °C	low · moderate · high · extreme
RH	0–100 %	very low · low · moderate · high
WS	0–100 km/h	calm · moderate · high · severe
FD	0–100 %	wet · moderate · dry · critical



Membership functions — temperature and fuel dryness shown; humidity and wind use the same four-set template. Outputs: fire risk index (4 sets), throttle urgency (3).

03 Rule base

4 inputs × 4 sets = 256 antecedent combinations, curated down to the 10 that are physically meaningful. **any** means the rule does not constrain that variable; min for AND, max for aggregation.

RULE	T	RH	WS	FD	RISK
R1	any	High	any	Wet	**Low**
R2	Low	Moderate	any	any	**Low**
R3	Moderate	any	any	Moderate	**Moderate**
R4	Moderate	any	High	Dry	**High**
R5	High	Low	any	any	**High**
R6	High	any	Severe	any	**Extreme**
R7	Extreme	Very Low	any	any	**Extreme**
R8	any	any	High	Critical	**Extreme**
R9	Extreme	any	any	Critical	**Extreme**
R10	any	High	Calm	any	**Low**

Firing strength α = min(μ of each antecedent); aggregated output μagg(y) = max over rules.

04 Defuzzification

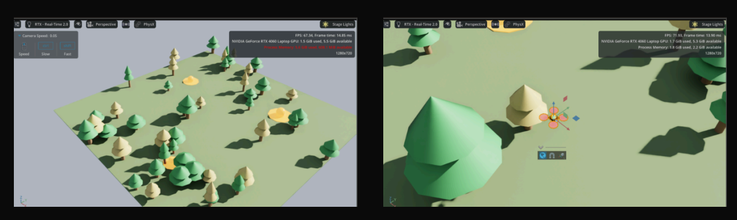
$$y^* = \int y \cdot \mu_{agg}(y)\, dy \,/\, \int \mu_{agg}(y)\, dy$$

Cost of the defuzzification step alone, median of 20,000 calls per scenario. The three methods do not separate on speed; they separate on continuity.

METHOD	COST	SMOOTHNESS	VERDICT
Centroid (COG)	2.5 µs	continuous	**selected**
Bisector	11.5–12.0 µs	continuous	viable
Mean of max (MOM)	2.7–2.9 µs	step jumps	rejected

MOM returns the mean of the aggregated set's maxima, so crossing a boundary jumps the output between peaks — a torque step on a 250 g airframe. Centroid integrates the whole area, so its output is **continuous in the input**.

05 Isaac Sim digital twin



65 m OpenUSD stage, 48 procedural pines, 3 active hotspots, chase and thermal cameras. Probe readings are taken from the stage every tick, so the controller is in the loop.

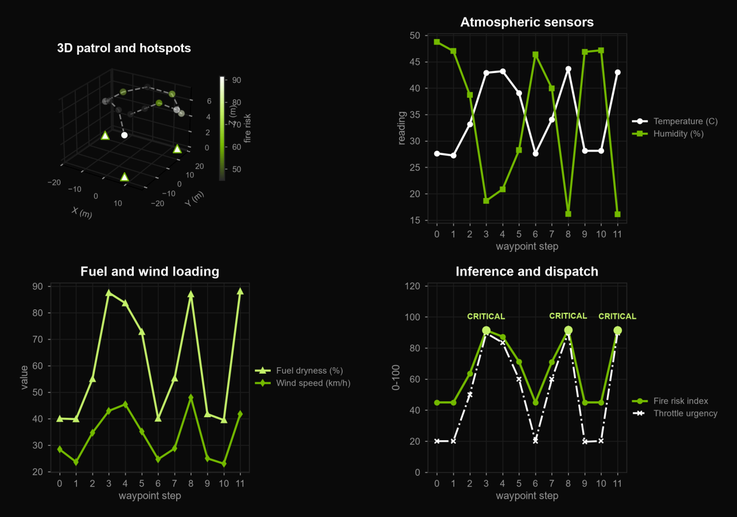
65 × 65 m procedural OpenUSD stage

3 hotspots, 44.5 °C cores

67–72 FPS, RTX Real-Time 2.0

0.078 ms median loop, p99 0.36 ms

06 Flight results



WP	TEMP	RH	FRI	THR.	ALERT
WP 00	27.9 °C	46.4 %	45.0	20.0 %	WATCH
WP 03	42.9 °C	18.1 %	91.4	89.6 %	**CRITICAL**
WP 06	28.2 °C	49.2 %	45.0	19.8 %	WATCH
WP 08	43.0 °C	17.9 %	91.5	89.6 %	**CRITICAL**

Twelve waypoints, no retuning. 89.6 % is a centroid output, not a hardcoded 90 %; recovery returns to 45.0 and 19.8 %, so nothing latches.

07 Conclusion

Four inputs, ten auditable rules and a closed loop inside Isaac Sim; **8/8 unit tests pass**. **Limitation:** simulation only, rule base expert-tuned, edge latency projected rather than measured. **Next:** ANFIS tuning, multi-UAV mesh, PX4 and Matrice 300 field trials.

Impact. Early thermal detection within the first 15–30 minutes of ignition catches fire before crown spread (SDG 13.1) and protects forest habitat (SDG 15.2); exact coordinates cut containment time, water use and crew exposure.

github.com/123haiku/csc3034-fuzzy-wildfire-sim
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